

GREG T. CHISM, PH.D.

 [Gchism94](#) •  [gregtchism.com](#) •  gchism@arizona.edu

EDUCATION University of Arizona, Graduate Interdisciplinary Program in Entomology and Insect Science
Ph.D. Entomology and Insect Science, 2022
Dissertation: The Influence of Nest Architecture on the Ant *Temnothorax rugatulus*
Advisor: [Dr. Anna Dornhaus](#)

University of California, Santa Barbara, Department of EEMB
B.S. Zoology, 2016
Advisors: Drs. Armand Kuris, Kevin Lafferty, Jonathan Pruitt

Shasta Community College
A.S. Biology, 2014

ACADEMIC Assistant Professor of Practice 2023 - Current
APPOINTMENTS College of Information Science

- Leading development of modular AI education platforms, including adaptive tutors and automated feedback systems.
- Developing and fostering AI education opportunities for students along with free computing resources for projects.

Computational & Data Science Educator (Data Scientist III) 2022 - 2023
Data Science Institute

- Organized events and workshops promoting data science education and open science practices.

CERTIFICATIONS

All of Us Certified Data User

- Registered Tier Certification (2024)

Carpentries Instructor Trainer (2023)

Carpentries Instructor (2022)

HONORS & Duke 3C Fellow, Cohort 5 2024 - Current
AWARDS

[Jetstream2 AI Pilot Fellow](#) (2024 - 2025): total award \$5,000

[Posit Table Contest - Best Individual Table](#) (2024)

EIS Carruth Award for Graduate Student Excellence (2021): \$500

GIDP - EIS Program Education Award (2020): \$250

OUTREACH & **Administrative Service (University of Arizona)**
SERVICE

1. Faculty Director of Instructional Computing 2025 - Current
College of Information Science

- Supervising [AI Core](#), a leading experiential learning program in AI + XR/VR under [Ash Black](#).
- Provide free computing resources to support scalable, open-source learning environments.

University Committees (University of Arizona)

1. Co-Lead: AI Roadmap Committees 2025 - 2026
Office for Responsible AI
 - Infrastructure & Resources; Research
 - Co-lead efforts to advance AI integration and leadership within the University of Arizona
2. Faculty Hiring Committee
 - CB2 Education Director / Faculty Hiring Committee 2024 - 2025

College Committees (University of Arizona - College of Information Science)

1. Faculty Awards Committee 2023 - Current
College of Information Science
 - Chaired committee 2025-2026.
 - Reviewed and assisted with faculty award nominations.
 - Helped to more than double the number of award nominations in 2024.
2. Faculty Hiring Committee
 - College of Info Sci - Science of Science Faculty Hiring Committee 2024 - 2025
3. Student Classroom Observer 2024
College of Information Science
 - Reviewed and recommended changes for online [ESOC 301: Qualitative Internet Research](#), Spring 2024.
 - Reviewed and recommended changes for in-person [ISTA 131: Dealing with Data](#), Fall 2024.

External Service (Arizona State University)

1. Quantum Computing Steering Committee (ASU - QCSC) 2025 - Current
 - Identified joint funding opportunities between ASU + UofA.

National/International Outreach

1. [Carpentries Instructor Trainer](#) Spring 2023 - Current
 - Train and mentor new instructors to promote inclusive teaching practices.

Other University Service

1. Co-Director, [Data Science Ambassadors Program](#) Spring 2024 - Current
 - Guide students in community-based data science initiatives and outreach projects.
2. Associate Member, DataLab 2023 - 2025
Data Science Institute
 - Collaborate on AI and data science workshops for students and researchers.
 - Develop just-in-time educational resources focused on computational tools and reproducible science.

Local/State Outreach

1. Chair/Deputy Chair, Research Bazaar Arizona 2023, 2022
 - Organized interdisciplinary workshops and expanded collaboration to three universities.
2. [Information on Tap](#): Election Integrity 2024
 - Discussed visualizing election uncertainty with the general Tucson public.
3. Industry Careers in Data Science Speaker Series 2022 - 2023
 - Developed and hosted a speaker series focused on academics transitioning to industry careers in data science.
 - Attracted ~20–40 attendees per session from diverse disciplines, career stages, and cultures.
4. [Insect Discovery Website](#) 2022
 - Designed content for the Insect Discovery website hosted by the U of A Extension Program.
5. Entomology Graduate Student Association (EGSA) Social Chair 2019 - 2020
 - Organized social events to encourage camaraderie and professional growth.
6. KEYS / SARSEF Research Internship Mentor 2018 - 2019
 - Mentored 4 Title I high school students in data science and research

PUBLICATIONS

Peer-reviewed Articles

(* indicates undergraduate students)

1. De La Rosa, J. S., **Chism, G. T.**, Herder, K. E., Mun, C. J., & Aaron, R. V. (2025). Do somatic symptoms bias depression screening? Reliability and equivalence of PHQ-8 in those with and without chronic pain: A nationally representative study of US adults *Journal of Affective Disorders*, 121496. [DOI](#)
2. **Chism, G. T.**, Nichols, W.* , and Dornhaus, A. (2024). Cavity geometry shapes overall ant colony organization through spatial limits, but workers maintain fidelity zones. *Animal Behaviour*, 216, 195-211. [DOI](#)
3. Swetnam, T. L., Antin, P. B., Bartelme, R., Bucksch, A., Camhy, D., **Chism, G.**, ... and Lyons, E. (2024). CyVerse: Cyberinfrastructure for open science. *PLOS Computational Biology*, 20(2), e1011270. [DOI](#)
4. McEwen, B. L., Lichtenstein, J. L., Fisher, D. N., Wright, C. M., **Chism, G. T.**, Pinter-Wollman, N., and Pruitt, J. N. (2020). Predictors of colony extinction vary by habitat type in social spiders. *Behavioral ecology and sociobiology*, 74, 1-9. [DOI](#)
5. Pruitt, J. N., Wright, C. M., Lichtenstein, J. L., **Chism, G. T.**, McEwen, B. L., Kamath, A., and Pinter-Wollman, N. (2018). Selection for collective aggressiveness favors social susceptibility in social spiders. *Current Biology*, 28(1), 100-105. [DOI](#)
6. Lichtenstein, J. L., **Chism, G. T.***, Kamath, A., and Pruitt, J. N. (2017). Intraindividual behavioral variability predicts foraging outcome in a beach-dwelling jumping spider. *Scientific Reports*, 7(1), 18063. [DOI](#)
7. Foster, W. C.* , Armstrong, C. M.* , **Chism, G. T.*** , and Pruitt, J. N. (2017). Smaller and bolder prey snails have higher survival in staged encounters with the sea star *Pisaster giganteus*. *Current Zoology*, 63(6), 633-638. [DOI](#)
 - Drafting a correction for misreported results (2024 - Ongoing)

Book Chapters

(^o indicates undergraduate students advisee)

1. Keiser, C. N., Lichtenstein, J. L. L., Wright, C. M., **Chism, G. T.**^{*}, Pruitt, J. N., Gonzalez-Santoyo, I., ... and Gonzalez-Tokman, D. (2018). Personality and behavioral syndromes in insects and spiders. *Insect behavior: From mechanisms to ecological and evolutionary consequences*, 236-256.

Works in Progress

(# indicates Open Access works, ^o indicates undergraduate advisee)

1. # De La Rosa, J. S., Herder, K.E.^o, Romero, R. D., Wolf, D. S. S., Largent-Milnes, T., Ibrahim, M. M., Pigott, S. S., **Chism, G. T.**, Meyerson, B. E., Pilitsis, J. G. and Brady, B. R., (2025). Impact of Chronic Pain on the Families of US Adults. *medRxiv*, 2025-02. [DOI](#)
 - In-Revision at PAIN Reports

Other Scholarship: Pre-Prints

(# indicates Open Access works, ^{*} indicates work done as a graduate student, ^o indicates undergraduate advisee)

1. # ^{*} **Chism, G. T.**, Faron, W.^o, and Dornhaus, A. (2022). *Temnothorax rugatulus* ants do not change their nest walls in response to environmental humidity. *bioRxiv*, 2022-06. [DOI](#)
2. # ^{*} Davis, S. M., **Chism, G. T.**, Maurer, M. M., Trejo, J. E., Garcia, R. J., and Schlenke, T. A. (2021). A hymenopteran odorant alerts flies to bury eggs. *bioRxiv*, 2021-09. [DOI](#)

Other Scholarship: Open Source Software

(# indicates Open Access works, ^{*} indicates work done as a graduate student, ^o indicates undergraduate advisee)

1. # ^{*} **Chism G. T.**, (2024). Gchism94/jupyterquest: v0.4.0 (v0.4.0). Zenodo. [DOI](#)
2. # Rice, L., Tate, S., Farynyk, D., Sun, J., **Chism, G.**, Charbonneau, D., Fasciano, T., Dornhaus, A. and Shin, M.C., (2020). ABCTracker: an easy-to-use, cloud-based application for tracking multiple objects. *arXiv preprint arXiv:2001.10072*. [DOI](#)

Other Scholarship: Open Source Code

(# indicates Open Access works, ^{*} indicates work done as a graduate student, ^o indicates undergraduate advisee)

1. # **Chism, G.**, & De La Rosa, J. (2025). MH-CP-UArizona/PHQ-MH-CP: (Pre-release) PHQ-MH-CP Compendium (v0.0.1). Zenodo. [DOI](#)
2. # ^{*} **Chism, G.**, Nichols, W.^o, and Dornhaus, A. (2023). NestArchOrg (v3.0.0). Zenodo. [DOI](#)
3. # ^{*} **Chism, G.**, Faron, W.^o, and Dornhaus, A. (2022). Gchism94/HumidityProject: Research Compendium for DOI: <https://doi.org/10.1101/2022.06.30.497551> (1.0.0). Zenodo. [DOI](#)
4. # ^{*} **Chism, G.** (2022). Gchism94/AntColonyPerformance: (Pre-release) Research Compendium for In preparation work (v0.1.0). Zenodo. [DOI](#)

Other Scholarship: Open Access Research Datasets

(# indicates Open Access works, * indicates undergraduate students)

1. # * **Chism, G.**, Nichols, W. *, and Dornhaus, A. (2022). Nest shape influences colony organization in ants: spatial distribution and connectedness of colony members differs from that predicted by random movement and is affected by nest space (1.0.0) [Data set]. Zenodo. [DOI](#)
2. # * **Chism, G.**, Faron, W. *, and Dornhaus, A. (2022). *Temnothorax rugatulus* ants do not change their nest walls in response to environmental humidity (1.0.0) [Data set]. Zenodo. [DOI](#)

Other Scholarship: Open Source Teaching Materials

(# indicates Open Access works)

1. # **Chism, G.**, Chen J. (2025). INFO-523-F25/INFO-523-F25: INFO 523 Fall 2025 Website (v1.0.0). Zenodo. [DOI](#)
2. # **Chism, G.** (2025). INFO-523-F25/INFO-523-F25: NFO-511-F25/info-511-f25: Fall 2025 Course Website + Materials (v1.0.0). Zenodo. [DOI](#)
3. # **Chism, G.** (2022). Data7 EDA In Python Book: Data7 EDA in Python Learning Materials (v2.0.0). Zenodo. [DOI](#) [DOI](#)
4. # **Chism, G.** (2022). Gchism94/Data7-rrtools-repro-research: Data7 Reproducible Research in GitHub and RStudio Workshop Series (v2.0.0). Zenodo. [DOI](#)
5. # **Chism, G.** (2022). Gchism94/Data7-EDA-In-R-Workshops: Data7 EDA in R Workshop Series (v1.0.0). Zenodo. [DOI](#)
6. # **Chism, G.** (2022). Gchism94/Data7-EDA-In-Shell: Data7 EDA in Unix Shell (v1.0.0). Zenodo. [DOI](#)
7. # **Chism, G.** (2022). Gchism94/Data7-EDA-In-SQL: Data7 EDA in SQL (v1.0.0). Zenodo. [DOI](#)
8. # **Chism, G.** (2022). Gchism94/DSI-KEYS2022-DataSci: Data7 KEYS 2022 Internship Data Science Materials (v2.0.0). Zenodo. [DOI](#)

MEDIA

1. **KJZZ Interview - Data centers, communities, and water** August 2025
2. **University of Arizona Graduate Center.** October 2024. *Elevating Graduate Research with Visual Communication: Insights from University Experts.*
3. **University of Arizona College of Information Science News.** February 2024. *Ensuring Creative Freedom: 8 Questions with Greg Chism...*
4. **University of Arizona News.** May 2019. *UA Students Earn NSF Graduate Research Fellowships.*
5. **University of Arizona News.** February 2019. *A Tasty Florida Butterfly Turns Sour.*

SCHOLARLY PRESENTATIONS

Invited Talks

1. **Chism, G.**
2. **Chism, G.** From Reproducibility to Reach - Building the AI Sandbox Through the Jetstream2 AI Fellowship. Jetstream2 AI Fellowship Cohort 2 Meeting. University of Indiana. Tucson, AZ. January 2025.

Invited Lectures

1. [Data Storytelling Masterclass, International Admissions](#) Spring 2025
2. History of Data Visualization Spring 2025
3. Data Storytelling Spring 2025

Title - MESA (Multidisciplinary Environment for Scientific Advancement)

- **Contribution:** 5% Credit, 0 FTE, PM-Effort: 1 Month, Sr. Personnel
- **PI:** Tyson Swetnam, University of Arizona
- **Funding Entity:** National Science Foundation (NSF) Integrated Data Systems & Services (IDSS)
- **Total award:** \$9,000,000 over 3-years
Direct costs - \$2,936,398 ; Indirect costs - \$1,628,629
- **Funding to UofA:** 78.32%; 5% F&A to College of Info Sci
- Submitted December 2025, **Not Awarded**

Title - MAESTRO: Modular AI Evaluation with Self-Testing and Resilient Oversight

- **Contribution:** 40% Credit, 0 FTE, PM-Effort: 1 Month, Lead-PI
- **Co-PIs:** Angela Cruze, University of Arizona; Tyson Swetnam, University of Arizona; Xiao Hu, University of Arizona
- **Funding Entity:** Office of Naval Research (ONR) Multidisciplinary University Research Initiatives (MURI) Program
- **Total award:** \$4,676,845 over 5-years
Direct costs - \$2,859,266; Indirect costs - \$1,586,442
- **Funding to UofA:** 100%; 60% F&A to College of Info Sci
- Submitted May 2025, **Not Awarded**

Title - Mental-HEALTH-MAP: Mental Health Equity through AI Learning, Tracking Health services, and Mapping Access Pathways for People with Persistent Pain.

- **Contribution:** 25% Credit, 0.1 FTE, PM-Effort: 1.25 Months, Co-Investigator
- **PI:** Jennifer De La Rosa, University of Arizona
- **Funding Entity:** National Institute of Health (NIH) -R34
- **Total award:** \$699,000 over 3-years
Direct costs - \$450,000; Indirect costs - \$249,000
- **Funding to UofA:** 100%; 25% F&A to College of Info Sci
- Submitted October 2024, **Not Awarded**

Title - University of Arizona Applied Research Corporation Response to SOO: "Data Management for the Directorate of Training and Doctrine (DOTD) Critical Task Site Selection Board (CTSSB) Process"

- **Contribution:** 38.07% Credit, 0 FTE, PM-Effort: 2.25 Months, Sr. Personnel
- **PI:** Angela Cruze, University of Arizona
- **Funding Entity:** United States Army Intelligence Center of Excellence (USAICoE)
- **Total award:** \$171,053 for 1-year
Direct costs - \$128,010; Indirect costs - \$43,043
- **Funding to UofA:** 100%; 35.07% F&A to College of Info Sci
- Submitted October 2024, **Not Awarded**

Title - AI Sandbox for Applied Neural Networks Education

- **Title:** AI Sandbox for Applied Neural Networks Education
- **Contribution:** 85% Credit, 0 FTE, Lead-PI
- **Co-PI:** Winslow Burleson, University of Arizona
- **Funding Entity:** National Science Foundation (NSF) National Artificial Intelligence Research Resource (NAIRR) Classroom Pilot; NAIRR250184
- **Total award:** 30K ACCESS GPU Credits for 1-year; \$50,000 in approximate value
- **Funding to UofA:** 100%; 0% F&A to College of Info Sci
- Submitted July 2025, **Awarded**

Title - Supporting Computational Resources for Advanced LLM and Data Science Modeling in a Capstone Course

- **Contribution:** 100% Credit, 0 FTE, Lead-PI
- **Funding Entity:** National Science Foundation (NSF) ACCESS Discover Award; SOC25004
- **Total award:** 1.5M NSF ACCESS Credits for 1-year; \$245,500 in approximate value
- **Funding to UofA:** 100%; 0% F&A to College of Info Sci
- Submitted January 2025, **Awarded**

Title - Benchmarking AI Models for Patient Journey Analysis Using MIMIC-III: Optimizing LSTM and Transformer Performance

- **Contribution:** 100% Credit, 0 FTE, Lead-PI
- **Funding Entity:** National Science Foundation (NSF) ACCESS Explore Award - MED250001
- **Total award:** 400k NSF ACCESS Credits for 1-year; \$72,802.50 in approximate value
- **Funding to UofA:** 100%; 0% F&A to College of Info Sci
- Submitted January 2025, **Awarded**